Fragile X syndrome
Kolkata data: 3.18% of patients tested; a test that exome can miss

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

Indian study (Kolkata): Bhowmik AD, Dutta S, Chatterjee A, Sinha S, Chattopadhyay A, Mukhopadhyay K. *Screening for Fragile X Syndrome among neurobehavioural patients from Kolkata, Eastern India.* J Clin Diagn Res 2009;3:1266-73. Outpatients of Manovikas Kendra, aged 1-35 years.

Who was tested

- 157 patients with neurobehavioural disorders (learning and behaviour problems)
- PCR sizing of CGG repeats, methylation-specific PCR, karyotype
- Authors suggest PCR to shortlist, Southern blot to confirm

What was found

- **5 of 157 (3.18%)** had Fragile X syndrome (the abstract text says four; the results body reports five)
- Psychiatric evaluation alone was not enough to recognise it
- Common CGG allele sizes differed from other populations

Other Indian data

Tamil Nadu (Nagarathinam et al., Advanced Genetics 2021): full mutation in 3 of 29 boys with intellectual disability (10.34%; small study). **Rural West Bengal (Dutta et al., IJMR 2009;130:714-9):** of 38,803 children surveyed, 179 with scholastic backwardness were tested and none had Fragile X. Lesson: yield depends on testing the right children.

The gene and the family

- **FMR1 CGG repeats:** normal up to 44; intermediate 45-54; premutation 55-200; full mutation over 200 (gene switched off: Fragile X syndrome).
- **Frequency:** about 1 in 4,000 to 6,250 males; females about half as often and usually milder.
- **Premutation carriers:** primary ovarian insufficiency before 40 in 12-28% of women (FXPOI); tremor-ataxia syndrome (FXTAS) in about 40% of men and 16-20% of women, usually from age 60-65.
- Prenatal testing in a later pregnancy: disease only; fetal sex never disclosed (PCPNDT).

EPISODE 15 · SALES FIELD SHEET

Fragile X: how to approach doctors

Topic	What to do or say
Doctors to visit	Paediatric neurologists, developmental paediatricians, child psychiatrists and autism centres, gynaecologists and IVF centres, neurologists.
Who to test	Unexplained intellectual disability or developmental delay; autism (boys and girls); family history of learning disability; menopause before 40; new tremor and imbalance after 50.
Opening question	"Doctor, when the exome is negative, do you also test Fragile X?"
Facts to quote	Kolkata clinic study: 5 of 157 (3.18%). Carriers: early menopause in 12-28% of women; tremor-ataxia in about 40% of men.
Test to suggest	Fragile X mutation analysis (FMR1 CGG repeats) by PCR; large expansions confirmed by Southern blot or methylation testing. In autism, microarray and Fragile X are the traditional first tests.
"Exome is already done"	"Exome can miss repeat expansions; Fragile X needs its own PCR test."
"Fragile X is rare"	"In a Kolkata clinic study it was 3.18%, and one result also guides the mother and relatives."
Sample	EDTA blood; counselling and written consent first. Confirm with the lab whether full mutations are confirmed in-house.
Never say	That a carrier is "a patient", or anything that blames the mother. Family testing is offered through a genetic counsellor.

References

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- Nagarathinam I et al. FMR1 gene CGG repeat distribution among three cohorts with intellectual disability, autism and primary ovarian insufficiency from Tamil Nadu. Advanced Genetics 2021;2(2). doaj.org
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- University of Washington Lab Medicine test guide: FMR1-related disorders. dlmp.uw.edu/test-guide/view/FMR1
- Hyman SL et al. Identification, evaluation, and management of children with autism spectrum disorder. Pediatrics 2020 (AAP clinical report).

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